

Support Vector Machines (scored task: 4 points)

Tasks (Lab 11):

1. Linear SVM and logistic regression.

Generate a two-dimensional linearly separable dataset using the following code. The dataset is constructed in such a way that each class contains a large Gaussian cluster and a small group of observations close to the separating boundary.

```
import numpy as np
import matplotlib.pyplot as plt

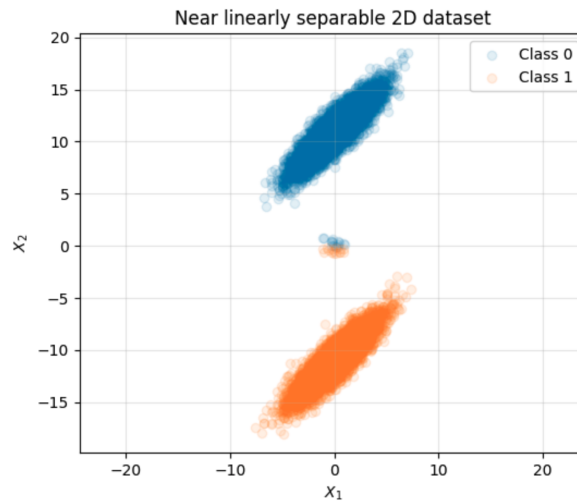
np.random.seed(42)

n_strip = 16
n_blob = 6000

# Class 0
x0_1 = np.hstack([
    np.random.normal(scale=0.6, size=(n_strip, 1)),
    np.random.uniform(0.3, 0.8, size=(n_strip, 1))
])
x0_2 = np.random.multivariate_normal(
    mean=[0, 10],
    cov=np.array([[4, 3.5], [3.5, 4]]),
    size=n_blob
)
X0 = np.vstack([x0_1, x0_2])
y0 = np.zeros(n_strip + n_blob)

# Class 1
x1_1 = np.hstack([
    np.random.normal(scale=0.6, size=(n_strip, 1)),
    np.random.uniform(-0.8, -0.3, size=(n_strip, 1))
])
x1_2 = np.random.multivariate_normal(
    mean=[0, -10],
    cov=np.array([[4, 3.5], [3.5, 4]]),
    size=n_blob
)
X1 = np.vstack([x1_1, x1_2])
y1 = np.ones(n_strip + n_blob)

X = np.vstack([X0, X1])
y = np.concatenate([y0, y1])
```



Before fitting the models, predict: will the SVM and logistic regression decision boundaries be similar or different? Which observations do you expect to influence each method most strongly?

Split the dataset into training and testing sets. Then fit two linear classifiers:

- logistic regression,
- linear SVM.

Before fitting the models, standardize the predictors.

For both methods:

- (a) compute training and testing accuracy,
- (b) draw the decision regions,
- (c) compare the obtained decision boundaries.

For the linear SVM additionally draw:

- the separating hyperplane,
- the margin lines,
- the support vectors.

Answer the following questions:

- (a) Both methods use linear decision boundaries. Why are the boundaries different?
- (b) Which observations become support vectors?
- (c) Are support vectors necessarily misclassified observations?
- (d) Does linear SVM estimate posterior probabilities $P(Y = 1 \mid X = x)$?
- (e) Why does the SVM boundary depend mainly on observations close to the separating boundary, while logistic regression is influenced by all observations?
- (f) How does the SVM boundary depend on the C parameter? Consider $C \in \{0.01, 0.1, 1, 10, 100\}$.
- (g) For each value of C , report the number of support vectors.

2. Kernel SVM.

Generate a two-dimensional nonlinear binary classification dataset using the following code.

```
from sklearn.datasets import make_moons

X, y = make_moons(
    n_samples=300,
    noise=0.25,
    random_state=123
)
```

Split the dataset into training and testing sets. Before fitting the models, standardize the predictors.

Fit the following classifiers:

- logistic regression,
- logistic regression with polynomial features of degree 2,
- logistic regression with polynomial features of degree 3,
- linear SVM,
- SVM with poly kernel (use $C = 10$, $\text{coef0} = 10$ and $d \in \{2, 3, 4\}$),
- SVM with RBF kernel.

For each method:

- (a) compute training and testing accuracy,
- (b) draw the decision regions,
- (c) compare the shape of the decision boundary.

For the SVM with RBF kernel fix $C = 1$. Fit the model for several values of the parameter **gamma**:

$$\gamma \in \{0.01, 0.1, 1, 10, 100\}.$$

For each value of **gamma**:

- (a) compute training and testing accuracy,
- (b) draw the decision regions,
- (c) compare the complexity of the decision boundary.

Answer the following questions:

- (a) Why do ordinary logistic regression and linear SVM perform poorly on this dataset?
- (b) Does the logistic regression with polynomial features of degree 3 produce a proper decision boundary for this task?
- (c) What does the RBF kernel allow the SVM to do?
- (d) What happens when **gamma** is very small?
- (e) What happens when **gamma** is very large?
- (f) Compare logistic regression with polynomial features and SVM with RBF kernel. In what sense are these approaches similar, and in what sense are they different?